

Qatar-2b

R-0786

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_240206 · created 2026-07-28T06:46:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460346.9598 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.219 +/- 0.065
Transit depth (Rp/Rs)^2	4.8 +/- 2.86 [%]
Orbital Inclination (inc)	85.79 +/- 1.74
Airmass coefficient 1 (a1)	1.33 +/- 0.76
Airmass coefficient 2 (a2)	-0.22 +/- 0.56
Scatter in the residuals of the lightcurve fit is	55.19 %
Best Comparison Star	#1 - [397, 99]
Optimal Aperture	2.88
Optimal Annulus	7.0
Transit Duration (day)	0.0734 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.219 ± 0.065 vs 0.16327 (deviation 0.79σ)
Duration fit vs published	0.0734 d vs 0.0767 d (deviation 0.39σ)
Mid-time O–C	-4.2 min
Depth SNR	3.1
Residual scatter	55.19 %

Light curve

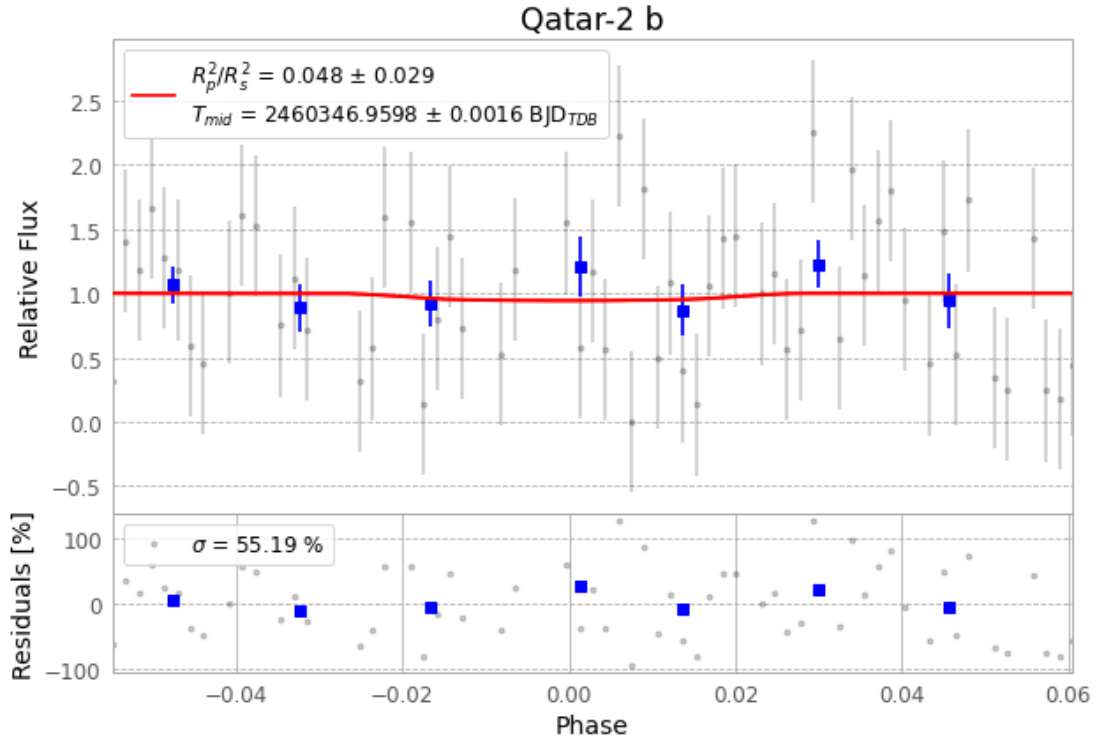


Figure 1 — FinalLightCurve_Qatar-2 b_2024-02-06.png (SHA-256 0985ed2f7aff78ea...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 159], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0d231fe2331d8d41...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

75 FITS frames (49 MB), Qatar-2240206091216.FITS → Qatar-2240206125415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-240206.log (abd31519867b...), FinalLightCurve_Qatar-2 b_2024-02-06.png (0985ed2f7aff...), FinalLightCurve_Qatar-2 b_2024-02-06.csv (9c1557e3ac09...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 06:46Z.